

CY 660: Photocatalysis: Fundamentals and Applications

Course Code : CY 660

Course Name : Photocatalysis: Fundamentals and Applications

L-T-P-C : 3-0-0-3

Prerequisites : CY 506 and CY 507 or equivalent/ Consent of instructor

Intended for : M.Sc. (Chemistry) and Ph.D.

Distribution : Elective PG and Ph.D

Approval: 14th Senate

Course Contents

- **Fundamental Concepts**

Photocatalyst introduction; light-matter interactions; semiconductor physics; quantum size effects; photoelectrochemistry and redox potentials; type I & II Semiconductors; Z-scheme photocatalytic systems. [8 Lectures]

- **Photochemical Processes**

Primary processes (homogeneous and heterogeneous) and time scales; natural photosynthesis (PSI & PSII); kinetics and reaction mechanisms; quantum yield and quantum efficiency; photonic yield and photonic efficiency; solar energy conversion efficiency; reactor design; reaction engineering. [8 Lectures]

- **Advanced Photocatalysts**

Selection of materials; band energy diagrams, band gap and band bending; pure and doped photocatalysts; composite and heterojunction photocatalysts; organic photocatalysts; dye-sensitized photocatalysts; strategies for development of efficient photocatalysts; review of current research. [10 Lectures]

- **Investigation Techniques**

X-ray diffraction; x-ray photoelectron spectroscopy; x-ray absorption spectroscopy; surface area measurements (BET analysis); photophysical measurements (UV-vis spectroscopy, Kubelka-Munk function for band gap determination, diffuse reflectance spectroscopy, fluorescence spectroscopy); Raman spectroscopy. [6 Lectures]

- **Technological Applications**

Degradation of organic contaminants in water; removal of metallic and inorganic pollutants; inactivation of microorganisms; artificial photosynthesis (water splitting and carbon dioxide reduction); organic transformations; biomass conversion; review of current research. [10 Lectures]

Text books:

1. Y. Nosaka, A. Nosaka, **Introduction to Photocatalysis: From Basic Science to Applications**, RSC, 2016.
2. R. Ameta, S. C. Ameta, **Photocatalysis: Principles and Applications**, CRC Press, 2016.



References:

1. J. Schneider, D. Bahnemann, J. Ye, G. L. Puma, D. D. Dionysiou (Eds.), **Photocatalysis: Fundamentals and Perspectives**, RSC Energy and Environment Series, 2016.
2. D. Dionysiou, G. L. Puma, J. Ye, J. Schneider, D. Bahnemann (Eds.), **Photocatalysis: Applications**, RSC Energy and Environment Series, 2016.